

Backend Engineering Assignment

Global Class Offering Booking System

You are building a simplified backend service for a global live-learning platform where teachers conduct online classes for students across different countries and timezones.

Each course/class can have multiple offerings (sections), and each offering contains multiple sessions.

Problem Statement

Examples

Example 1 — Weekly Course

An 8-week course may have:

- Every Saturday, 6 PM–7 PM for 8 weeks

Example 2 — Summer Camp

A 5-day summer camp may have:

- Monday–Friday, 5 PM–6 PM during the same week

Teachers create offerings in their own timezone, while parents/students should view schedules in their local timezone.

Your task is to design and implement a production-ready backend service for this workflow.

Core Concepts

1. Course/Class

Examples:

- Python Coding
 - Art Drawing Class
 - Public Speaking
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2. Offering / Section

A schedulable version of a course.

Examples:

- Saturday Batch
- Weekday Summer Camp
- Evening Batch

Each offering contains multiple sessions.

3. Sessions

Actual meeting times belonging to an offering.

Example

Offering:

- Minecraft Coding — Saturday Batch

Sessions:

- June 6 → 6 PM–7 PM
 - June 13 → 6 PM–7 PM
 - June 20 → 6 PM–7 PM
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Functional Requirements

1. Teacher APIs

Teachers should be able to:

- Create an offering

- Add sessions to the offering
- View upcoming offerings and sessions

Each session should contain:

- Offering ID
 - Teacher ID
 - Start time
 - End time
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2. Parent APIs

Parents/students should be able to:

- View available offerings
- Book an offering
- View booked offerings

Important

Parents may belong to a different timezone than the teacher.

Session timings shown to parents should appear correctly in the parent's local timezone.

Booking Rules

Rule 1 — Booking Happens at Offering Level

Parents book the entire offering, not individual sessions.

Example

Booking an 8-week course means booking all 8 sessions together.

Rule 2 — Time Conflict Locking

Once a parent books an offering, all session times belonging to that offering should become locked for that parent.

The parent should NOT be able to book another offering that overlaps with ANY session timing already booked.

Example

Parent books:

- Minecraft Coding (Saturday Batch)
- June 7 → 5 PM–6 PM
- June 14 → 5 PM–6 PM
- June 21 → 5 PM–6 PM

Then the parent should NOT be able to book:

- Roblox Game Design
- June 14 → 5:30 PM–6:30 PM

because one or more sessions overlap.

Rule 3 — Concurrent Booking Handling

The system should correctly handle simultaneous booking attempts.

Examples

- Multiple parents booking the same offering simultaneously
- A parent attempting overlapping bookings through multiple requests

Your solution should ensure:

- Data consistency
 - Prevention of invalid bookings
 - Proper concurrency handling
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Technical Requirements

You may use:

Preferred

- Java + Spring Boot

OR

- Node.js / NestJS

Database

- PostgreSQL
 - OR MySQL
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Expected APIs

Teacher APIs

- Create offering
- Add sessions to offering
- Get teacher offerings

Parent APIs

- Get available offerings
 - Book offering
 - Get bookings
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Expectations

We are not looking for a very large project.

We care more about:

- Clean backend design
 - Correct handling of concurrency
 - Good database design
 - Proper timezone handling
 - Thoughtful error handling
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Evaluation Criteria

Your solution will be evaluated on:

- API correctness
- Database design
- Code structure and readability
- Handling of simultaneous booking requests
- Timezone handling
- Conflict detection logic
- Error handling and validations
- Overall engineering approach

Submission Guidelines -

https://docs.google.com/document/d/17Vxr4twczr3WwOqeyzjLBe8s-Wmylrl_a89-Eif4nX4/edit?tab=t.0